Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Fairfax, Virginia

PROFESSIONAL SUMMARY

ML and platform engineer with 6+ years building production Python services, ML pipelines, and data infrastructure. At Intuit, building and deploying ML pipelines (scikit-learn, XGBoost) on AWS with CI/CD automation, model performance monitoring, and evaluation harnesses to catch regressions before release. Built AgentProbe -- a developer platform toolkit for LLM evaluation with automated CI regression tests, model performance scoring across five metrics, and multi-provider routing. Strong background in Docker, Kubernetes, Terraform, Prometheus, Grafana, GitHub Actions CI/CD, and AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

ML Pipelines & Deployment | Model Monitoring & Drift | CI/CD Automation | Python (6+ yrs) | PyTorch & scikit-learn | Docker & Kubernetes | AWS / Azure / GCP | Terraform / IaC | Prometheus & Grafana | Developer Enablement | Generative AI & LLMs | Production Systems

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Build and maintain **ML pipelines** (scikit-learn, XGBoost) covering data ingestion, feature engineering, model training, evaluation, and prediction API deployment on AWS.
- Monitor **model performance, data quality, and pipeline health**; implement evaluation and regression tests (LLM-as-judge and golden datasets) to detect regressions before production release.
- Integrate Anthropic Claude into Python/FastAPI services with structured-output schemas, input validation, and guardrails; add CI/CD regression tests to catch LLM quality drops automatically.
- Develop backend APIs and data pipelines with React/TypeScript front ends; containerize with Docker and ship through **GitHub Actions CI/CD** to AWS (S3, EC2, Lambda, API Gateway) with logging and observability.
- Build internal tools for finance teams -- natural-language query interfaces and automated reporting pipelines reducing manual workload for non-engineers.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant SaaS, 100K+ monthly active users

- Delivered 20+ features end-to-end (Python, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints on a high-scale production platform; owned reliability and performance of deployed services.
- Built a **RabbitMQ-based async processing system** for high-volume data workflows across thousands of tenant records, with retry logic and idempotency for reliable pipeline execution.
- Optimized 200+ SQL queries using execution plans and indexing (30% performance gain); owned production data debugging when pipelines produced incorrect output.
- Built Python/Bash **automation workflows** for operational teams (80% manual workload eliminated); internal tooling reducing incident triage time by 40%.
- **AWS S3 migration** with Libsodium encryption for 100% PHI compliance; Docker microservices redesign (50% better scalability, 30% faster deployments); Terraform for infrastructure management.
- Reviewed 300+ pull requests; mentored junior engineers on production-quality coding practices.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built data pipelines and application modules (Python, Java, PostgreSQL) for large-scale data import and reconciliation (1.6M records); led AWS S3 cloud migration from legacy storage.

PROJECTS

AgentProbe -- LLM Evaluation Platform & Developer Toolkit

Live on GitHub

Built a **developer enablement platform** for LLM evaluation: 15-endpoint FastAPI backend, multi-provider routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama), and a 5-metric model performance scoring engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance). **Automated CI regression pipeline** (GitHub Actions) runs model performance checks on every commit using curated golden datasets with train/test/validate splits -- alerts when metrics degrade.

Python · FastAPI · LangGraph · ChromaDB · PyTorch · GitHub Actions CI/CD · Anthropic · OpenAI · Azure OpenAI -- github.com/venkatalakshmiparimala/AgentProbe

Leasel -- Production ML Pipeline with Evaluation Harness

Built an end-to-end **ML pipeline**: OCR ingestion, feature extraction, hybrid retrieval (TF-IDF + semantic reranking), 8-agent LangGraph orchestration, and CI evaluation harness. **Model performance monitoring** via LLM-as-judge: precision 93.6%, F1 87.9% across 1,240 test cases with automated regression checks on every build.

Python · LangGraph · FastAPI · scikit-learn (TF-IDF) · OCR · SQLite · GitHub Actions CI/CD -- github.com/venkatalakshmiparimala/Lease_complaince_rag

CrashClarity -- ML Model Training & Deployment Pipeline

Built a full ML pipeline on 104K+ UK accident records: data ingestion, feature engineering, model training (Random Forest + Logistic Regression), evaluation (85%+ accuracy), and deployment to an interactive GitHub Pages dashboard.

Python · R · scikit-learn · Random Forest · Logistic Regression · GitHub Pages

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

TECHNICAL SKILLS

ML & AI: Python, PyTorch, scikit-learn, XGBoost, PySpark, Pandas, NumPy, LLMs (Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama), LangGraph, LangChain, RAG Pipelines, Embeddings, BERT, spaCy, HuggingFace

ML Pipelines & Monitoring: ML pipeline automation, model deployment, model performance monitoring, data drift detection, CI/CD regression testing, LLM-as-Judge evaluation, GitHub Actions, Jenkins CI/CD

Observability: Prometheus, Grafana, OpenTelemetry, AWS CloudWatch

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, Ansible

Backend & Data: FastAPI, Flask, Node.js, Java, Spring Boot, TypeScript, React, PostgreSQL, MySQL, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Query Optimization, RabbitMQ, Kafka, Celery

Practices: REST APIs, Microservices, Agile/Scrum, TDD, Code Review, HIPAA/PHI Compliance, Developer Enablement, Self-service Platform Tooling